

Class 12: Inverse of a Matrix

Definition: A square matrix A has an inverse matrix if

Example: Find the inverse of the matrix $A = \begin{bmatrix} 1 & -1 \\ 3 & -2 \end{bmatrix}$

We need a matrix $\begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix}$ such that

So we need to solve two matrix equations

Two questions:

1. Which matrices are invertible?
2. How do you find the inverse when it exists?

How to:

To find the inverse of a square matrix A you augment A by I and row reduce.

The answer to the first question is also revealed by this how to process.

What does this tell us about the solution to the matrix equation $A\mathbf{x} = \mathbf{b}$?

Elementary Matrices:

When row reducing a matrix A we can use three types of operations. Row replacement; Row switch; and Row multiple. Each of these operations on A can be accomplished by multiplying the matrix A by the appropriate elementary matrix E .

Examples:

Replace R_2 by $R_2 + 2R_1$

Switch R_2 and R_3

Multiply R_1 by -1

Imagine it takes p row operations to reduce A to the identity matrix. That means we multiply A by p elementary matrices. We have the equation

$$I = E_p \cdots E_2 E_1 A = (E_p \cdots E_2 E_1) A$$

A^{-1} is a product of elementary matrices.

Class Investigation

Use a calculator to find the inverse of the following matrices and try to guess a formula for the inverse of the $n \times n$ matrix with the same form.

	A	A^{-1}
$n=3$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	
$n=4$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	
$n=n$		

Discover a formula for the inverse of a diagonal matrix

$$\begin{bmatrix} a_{11} & 0 \dots & 0 \\ 0 & a_{22} & 0 \\ \vdots & \vdots & \vdots \\ 0 & 0 \dots & a_{nn} \end{bmatrix}^{-1} =$$